

Can you read all these high-level theory descriptions, and cluster them into similar theories, while describing what the clusters mean?
(The theories are found at:
<https://peterajansen.github.io/theories-onco2-june8-2026/json/theories.json>)

I'll pull the JSON directly, inspect its structure, then cluster by recurring constructs/claims rather than just surface keywords. I'll keep the clusters interpretable and flag any singleton or ambiguous theories rather than forcing them into bad groups.

I've got the top-level structure now. The theories are heavily overlapping: many are different framings of the same few ideas, so I'm clustering by the main causal/explanatory claim rather than by title similarity alone.

Thought for 1m 53s

I read the JSON and treated the theories as **overlapping conceptual variants**, not mutually exclusive bins. The cleanest clustering is around **what the theory thinks the “driver-negative” label is hiding**: missed lesions, pathway-state convergence, regulatory/protein-layer activation, lineage context, or ecosystem/state-first biology.

High-level cluster map

Cluster	Main idea	Theories	What the cluster means
1. Assay-compression / hidden-lesion recovery	“Driver-negative” is often an artifact of limited testing.	1843, 1849, 1852, 1855, 1863, 1865, 1869, 1872, 1875	These theories argue that many tumors are only negative relative to the assay used. Broader DNA/RNA/WGS/fusion/splice/copy-number/regulatory assays “decompress” the negative bucket into rare but real drivers. This is the most lesion-first cluster. The JSON repeatedly frames driver negativity as assay-relative and sample-quality dependent. peterajansen.git... +2
2. Pathway-output / state-convergence theories	Different upstream events converge onto a small number of functional states.	1841, 1844, 1845, 1847, 1853, 1857, 1873, 1874, 1876, 1878, 1879, 1880, 1881, 1882	These are the central “state-first” or “output-first” theories. They say the important object is not the individual rare lesion but the recurrent output: MAPK/PI3K signaling, proliferation/checkpoint addiction, immune/stromal states, lineage/metabolic programs, etc. This is probably the dominant family in the set. peterajansen.git... +3
3. Regulatory / structural / proteogenomic hidden-driver theories	The driver is real, but it lives outside coding-hotspot logic.	1842, 1846, 1854, 1868, 1871	These theories focus on enhancer hijacking, super-enhancer/SV rewiring, promoter lesions, exon skipping, ligand presentation, and phosphoproteomic activation. They are midway between “hidden lesion” and “state” theories: the cause may be a structural/regulatory event, but the evidence may appear as expression or phosphosite state. peterajansen.git... +2
4. Ligand-and-lineage substitution theories	Canonical kinase activation is replaced by ligand context or lineage-state context.	1850, 1851, 1866, 1867, 1870, 1883	These theories emphasize mucinous/gastric/NKX2-1-low biology, NRG1/ERBB ligand-style signaling, lineage-factor changes, and the idea that epithelial identity determines which hidden driver architectures are possible. In this cluster, lineage is not just a phenotype; it shapes the causal mechanism. peterajansen.git... +3
5. Ecosystem / residual state-first disease	After hidden drivers are removed, some tumors are maintained by tumor-state or microenvironment-state programs.	1848, 1858, 1859, 1864, 1877, 1883	These theories are the strongest alternative to the “we just missed a rare gene” view. They argue that the residual class may be driven by angiogenic/stromal niches, immune topology, fibroblast-rich states, glycolysis/MIF-HIF1α, ferroptosis resistance, estrogen/SRC signaling, KAC-like precursor states, or quiescent/stem-like states. peterajansen.git... +3
6. Distributed weak-driver / suppressor-loss convergence	No single classic oncogene is necessary; combinations of weaker lesions reinstate pathway output.	1860, 1876, 1882	These theories are more specific than the broad state-convergence cluster. They focus on cooperative loss of negative regulators, especially NF1/RASA1/PTEN-like logic, and on distributed subthreshold alterations that collectively restore RAS/PI3K/MAPK output. peterajansen.git... +2
7. Mechanism-ontology / cross-cancer architecture theories	The field should organize hidden drivers by mechanism families, not long lists of genes.	1856, 1865, 1873	These are meta-theories. They compress many findings into a reusable ontology: kinase fusions, ligand fusions, splice-mediated receptor stabilization, enhancer/SV rewiring, suppressor-loss convergence, protein-state activation, and ecosystem substitution. They are useful for paper framing because they turn scattered rare events into a finite discovery grammar. peterajansen.git... +2

What the clusters mean conceptually

The theories mostly differ along two axes.

First, they differ on whether **driver-negative LUAD is mainly a diagnostic-resolution problem** or a **true biology problem**. The assay-compression cluster says many cases are not biologically negative; they are just negative under limited clinical testing. The residual state-first and ecosystem clusters say that even after rescue, some tumors may not have a single dominant classic genomic driver.

Second, they differ on the **causal layer** where the hidden driver lives: coding DNA, RNA splicing/fusions, structural/regulatory genome architecture, phosphoproteomic signaling, lineage/chromatin state, or tumor microenvironment. The strongest synthesis across the set is: **driver-negative LUAD is probably not one category; it is a mixture of assay-missed canonical disease, noncanonical pathway-reinstatement disease, and true residual state/ecosystem disease.**

My suggested “merged” taxonomy

For a paper or research program, I would collapse the 40-ish theories into this simpler taxonomy:

1. Pseudo-negative / assay-limited LUAD

Cases rescued by better testing: hidden EGFR/KRAS, fusions, MET exon 14, NTRK/RET/ROS1, rare RAF/RAS-family alterations, amplifications.

2. Noncanonical pathway-reinstated LUAD

Cases without a classic driver but with restored MAPK/PI3K/RTK output through suppressor loss, regulatory rewiring, splice events, ligand activation, or phosphoproteomic activation.

3. Lineage-context hidden-driver LUAD

Cases where mucinous/gastric/NKX2-1-low/KAC-like/never-smoker lineage context changes which drivers are possible and how signaling is interpreted.

4. Ecosystem/state-maintained residual LUAD

Cases where the effective driver is a stable tumor-state or microenvironmental attractor: immune-cold, angiogenic-stromal, fibroblast-rich, ferroptosis-resistant, glycolytic, estrogen/SRC, stem-like/quiescent.

5. Therapeutic output-state classes

A practical layer cutting across all of the above: classify by actionable output—MAPK/PI3K-active, checkpoint/proliferative, immune-hot/cold, angiogenic/CAF-rich, lineage-plastic, metabolic.

Most redundant groups

The biggest redundancy is among the **state-convergence/output-first** theories: 1841, 1844, 1845, 1847, 1853, 1857, 1874, 1878, 1879, 1880, 1881, and 1882 are mostly variants of the same core claim. They differ in emphasis, but the shared thesis is: **negative by genomics does not mean negative by pathway state.**

The second-largest redundancy is among the **assay-compression / hidden-driver architecture** theories: 1843, 1849, 1852, 1855, 1863, 1865, 1869, 1872, and 1875. These are mostly alternate formulations of: **the “negative” bin decomposes as resolution improves.**

The most distinct theories are the ones that push beyond missed-lesion logic: **1848, 1858, 1859, 1864, 1877, and 1883**, because they make the stronger claim that some residual tumors may be genuinely **state-first/ecosystem-first** rather than merely under-assayed.

      Sources

Thought for 34s >

They live primarily in the group I called:

Distributed weak-driver / suppressor-loss convergence

That is the “brakes are broken” group. The core idea is **not** that there is one missed activating oncogene, but that loss of negative regulators removes restraints on signaling until the tumor recreates a driver-like output state.

The most direct match is **theory-1876: “Substitute-Driver Convergence Theory for Oncogene-Negative LUAD.”** It explicitly says that “combinatorial loss of negative regulators” can substitute for a canonical oncogene by jointly activating **RAS/MAPK** and **PI3K/AKT**, and that the operative driver unit is **pathway-state sufficiency, not lesion identity.** peterajansen.git...

The key law inside it is:

Suppressor-convergence substitute-driver law

This law says that in LUADs lacking canonical proto-oncogene activation, concurrent inactivation of negative regulators across RAS and PI3K pathways can be sufficient to initiate and maintain malignancy. It names **NF1 + RASA1 + PTEN loss** as the validated substitute-driver configuration, with **RASA1 + NF1 co-loss** as a human clinical analogue. peterajansen.git...

So, in my taxonomy, these are not mainly in the “hidden rare driver” bucket. They are in a **substitute-driver / pathway-brake-loss** bucket.

Closest specific theories/laws

Theory	Law / statement	How it relates
theory-1876 — Substitute-Driver Convergence Theory	Suppressor-convergence substitute-driver law	The cleanest “brakes are broken” formulation: NF1/RASA1/PTEN or RASA1/NF1 loss removes pathway restraint and substitutes for a canonical oncogene. peterajansen.git...
theory-1876	Dual-pathway therapy law	Therapeutic corollary: if the tumor is maintained by convergent RAS/MAPK + PI3K/AKT activation, combined blockade of upstream RAS-input and PI3K/AKT-output nodes should outperform monotherapy. peterajansen.git...
theory-1878 — Convergent Pathway-State Theory	Convergent-pathway substitution law	Broader version: tumorigenesis can be sustained by combinations of non-oncogene alterations that recreate downstream RAS/MAPK and PI3K/AKT/mTOR outputs. This includes tumor-suppressor co-loss as one major mechanism. peterajansen.git...
theory-1841 — Convergent State-Driver Theory	Driver-negativity as hidden pathway convergence	Places suppressor loss as one mechanism inside a larger state-convergence theory: “derepression of RAS/PI3K by loss of negative regulators” is one route to driver-negative LUAD. peterajansen.git... +1
theory-1875 — Hidden Lesion Hierarchy Theory	Hidden lesion hierarchy law	Puts suppressor-loss biology at the deepest/residual end of the hierarchy: after rare coding drivers, fusions, SVs, and enhancer events are removed, the remaining class is enriched for tumor-suppressor co-loss or unresolved pathway-state biology. peterajansen.git... +1
theory-1867 — Lineage-State Diversification Theory	Suppressor-loss/proteomic-state class	Treats suppressor-loss as one of several residual attractors, specifically a class where pathway activation comes from loss of negative regulators, phospho-network rewiring, or epigenetic/protein-level inactivation rather than one dominant oncogene. peterajansen.git...

Where it does not primarily belong

I would **not** primarily place these in the **ecosystem/state-maintained residual LUAD** group. theory-1877 explicitly says that ecosystem-state theories do **not** directly explain NF1/RASA1/PTEN or RASA1/NF1 co-loss tumors, because those are dominated by dual RAS/PI3K signaling rather than stromal ecosystem biology. [peterajansen.git...](#)

Best label for the group

I'd rename the group from my earlier answer to something more precise:

Suppressor-loss substitute-driver / pathway-brake-loss theories

One-sentence meaning:

These theories propose that some “driver-negative” LUADs are not missing an accelerator; instead, they have lost enough brakes—NF1, RASA1, PTEN, STK11/KEAP1-like restraints, or related pathway suppressors—that canonical growth pathways become active anyway.

      [Sources](#)

Thanks. Are there any obvious wins/information gains/clear utility from any of the theories in this file, relative to current knowledge? For the most impactful ones, can they be tested using simple, low-cost, straightforward/easy-to-implement fully computational experiments, that would be suitable for an interdisciplinary masters student studying NLP and bioinformatics? (Please make reference to specific theory IDs)

I'll treat “utility” as: a theory adds a practical distinction beyond standard “sequence more / profile more” advice, and can be tested from public LUAD multi-omics or text-mined literature without wet-lab work.

One emerging pattern: the best “wins” are not new rare-gene hunts. They're computational triage frameworks that turn “driver-negative” from a leftover label into measurable subtypes, which is much more testable.

Thought for 1m 22s ›

Yes. The **clearest wins** are theories that convert “driver-negative LUAD” from a vague residual label into **testable, computable strata**. The most useful ones are not necessarily the most biologically exotic; they are the ones that give you a measurable classifier, an enrichment test, or an information-gain comparison.

Bottom line

The highest-utility theories are:

1. **theory-1841 / theory-1881:** state-space classification beats “driver-negative” as one undifferentiated bin.
2. **theory-1875:** driver-negativity is assay-depth-dependent; model the negative pool as a shrinking hierarchy.
3. **theory-1876:** some tumors are “brake-loss” / suppressor-convergence tumors, especially NF1/RASA1/PTEN-like cases.
4. **theory-1846:** phosphoproteomic/protein outliers reveal hidden dependencies missed by DNA.
5. **theory-1842 / theory-1856:** hidden drivers fall into recurring mechanistic families: ligand fusion, splice/turnover rewiring, enhancer/SV rewiring, suppressor-loss, phospho-state activation.
6. **theory-1883 / theory-1874:** the residual class after hidden drivers and suppressor-loss may be state-first, lineage/metabolic/ecosystem-driven disease.

The practical opportunity is: **build a computational framework that assigns driver-negative LUADs to interpretable mechanism/state classes, then show that those classes explain more biology than the binary “driver-negative” label.**

1. Most obvious win: state-first classification of driver-negative LUAD

Relevant theories/laws:

theory-1841, especially the claim that driver-negativity often reflects hidden pathway convergence rather than pathway absence; and its prediction that state-first labels such as RAS/PI3K-active, angiogenic-stromal, immune-high, proliferation/checkpoint, lineage-reprogrammed, and metabolism-rewired should predict outcomes better than the binary label “driver-negative.” The theory explicitly says state-defined grouping should be tested against lesion-defined grouping and that lack of reproducible state classes would weaken the theory. Peter A Jansen

Why this is useful relative to current knowledge:

The individual pieces already exist—hidden fusions, rare mutations, proteogenomic subtypes, immune states, lineage states—but the information gain is in **collapsing many partial subtype systems into one practical state-space**. The theory itself says the novelty is the integrated claim that these findings collapse into recurring state classes more useful than the residual label “driver-negative.” Peter A Jansen

Student-suitable computational experiment:

Use TCGA-LUAD and CPTAC-LUAD to define a canonical-driver-negative subset, then compute state scores from RNA/protein/phosphoprotein data:

- RAS/MAPK and PI3K/AKT activity
- proliferation/checkpoint score
- angiogenesis/CAF/stromal score
- immune-hot/immune-cold score
- NKX2-1/HNF4A/mucinous/lineage score
- glycolysis/OXPHOS/ferroptosis/stress score

Then test whether these state labels improve prediction of survival, recurrence, immune subtype, or proteogenomic subtype compared with a binary driver-negative label. TCGA-LUAD data are available through the GDC, and CPTAC is explicitly designed to add proteomic and proteogenomic layers to cancer genomics. GDC Portal +1

Expected result if theory is right:

Driver-negative tumors will not form one noisy residual group. They will cluster into a few reproducible state classes, and those classes will explain survival/prognosis/pathway activity better than “driver-negative yes/no.”

Difficulty: low to moderate. Mostly Python/R, public data, clustering, survival models, gene-set scoring.

2. Very clean win: assay-depth rescue hierarchy

Relevant theories/laws:

theory-1875, especially the **Assay-depth rescue law** and **Hidden lesion hierarchy** framing. The theory says apparent driver-negative LUAD should shrink as testing expands from hotspot panels to broad DNA/RNA/structural/noncoding assays, and that the deepest residual class should become enriched for suppressor co-loss and state biology. Peter A Jansen

There is also another assay-depth formulation in the file: the rescue fraction should decline as the initial negative definition becomes broader, more orthogonal, and more purity-aware. Peter A Jansen

Why this is useful relative to current knowledge:

This is not “discover another rare gene.” It gives a **measurement model** for driver-negativity. A tumor labeled negative after EGFR/KRAS/ALK testing is not equivalent to a tumor labeled negative after WGS + RNA fusion calling + phosphoproteomics. That distinction is obvious once stated, but often gets blurred in papers and cohorts.

Student-suitable computational/NLP experiment:

Do a literature-mining meta-analysis:

Extract from papers:

- cohort size
- initial negative definition: EGFR-only, EGFR/ALK, EGFR/KRAS/ALK, targeted panel, WES, WGS, RNA-seq, etc.
- assay modalities used for reanalysis
- number/fraction rescued
- rescued lesion class: rare coding variant, fusion, MET exon 14, amplification, SV/noncoding, suppressor-loss, protein-state

Then fit a simple model:

```
rescue_fraction ~ initial_assay_depth + RNA_used + WGS_used + tumor_purity_filter + histology +
```

Expected result if theory is right:

Rescue fraction should be high in weakly screened cohorts and lower in already deeply screened cohorts. The residual cases after deeper screening should be more enriched for suppressor-loss, regulatory, phosphoproteomic, lineage, or ecosystem explanations.

Difficulty: low. This is ideal for an NLP + bioinformatics student: document retrieval, schema extraction, manual validation on 30–50 papers, then regression/meta-analysis.

High value: very high, because it clarifies a major confounder in the whole research area.

3. Strong mechanistic win: suppressor-loss / “broken brakes” substitute-driver class

Relevant theories/laws:

theory-1876, especially the **Suppressor-convergence substitute-driver law**. It states that in LUADs lacking canonical proto-oncogene activation, concurrent inactivation of negative regulators across RAS and PI3K pathways can be sufficient to initiate and sustain malignant growth, with **NF1 + RASA1 + PTEN loss** as the validated configuration and **RASA1 + NF1 co-loss** as a human analogue. Peter A Jansen

The broader version appears in theory-1876’s description: pathway-state sufficiency, not lesion identity, is the operative driver unit. Peter A Jansen

Why this is useful relative to current knowledge:

This is probably the cleanest biologically actionable theory in the file. It reframes a subset of driver-negative tumors as **not missing an accelerator**, but having lost enough brakes to recreate RAS/MAPK and PI3K/AKT signaling.

Student-suitable computational experiment:

Use TCGA-LUAD plus CPTAC-LUAD:

1. Define canonical-driver-negative or driver-poor tumors.
2. Score **NF1**, **RASA1**, **PTEN** disruption using mutation, copy loss, expression loss, methylation if available, and protein loss if CPTAC is used.
3. Compute MAPK and PI3K/AKT output scores from RNA/protein/phosphoprotein data.
4. Test whether NF1/RASA1/PTEN-disrupted tumors are enriched for high MAPK + high PI3K output.
5. Compare against canonical KRAS/EGFR tumors and against truly quiet negatives.

Expected result if theory is right:

A subset of driver-negative tumors should show dual pathway activation and should be statistically enriched for negative-regulator loss.

Difficulty: moderate. The mutation/CNA/RNA version is straightforward; the phosphoproteomic version is stronger but requires more data wrangling.

Why it is publishable-ish:

It gives a crisp subtype with a falsifiable enrichment test. It is much less vague than “maybe there are other pathways.”

4. High-value multiomics win: phosphoproteomic output beats genotype alone

Relevant theories/laws:

theory-1846, especially the law that **protein/phosphoprotein outliers reveal actionable hidden dependence missed by genotype alone**. The theory says unresolved LUAD should be interrogated as an output system where kinase-node phosphorylation, activated receptor signaling, enhancer hijacking, or ligand-receptor loops reveal targetable dependence even when canonical genomic lesions are absent. Peter A Jansen

Why this is useful relative to current knowledge:

CPTAC already exists, and proteogenomics is already a known direction. The win is to make this **driver-negative-specific**: among tumors where DNA is unhelpful, does protein/phospho state recover coherent biology?

CPTAC-LUAD includes proteome and phosphoproteome data from a LUAD discovery cohort; the OmicsDI record describes 111 tumors, 102 paired normals, and global proteome/phosphoproteome analysis. OmicsDI

Student-suitable computational experiment:

In CPTAC-LUAD:

- Identify canonical-driver-negative or driver-poor tumors.
- Compute phosphosite/protein outlier scores.
- Ask whether DNA-negative tumors with phospho-output positivity cluster near known EGFR/KRAS/ALK-like pathway states.
- Compare models:

```
Model A: mutation/CNA only
Model B: mutation/CNA + RNA state
Model C: mutation/CNA + RNA + protein/phosphoprotein outliers
```



Evaluate information gain for subtype prediction, pathway activity, survival, or inferred drug vulnerability.

Expected result if theory is right:

Some DNA-negative tumors will be “output-positive” at the protein/phosphosite level, and DNA+phospho models will explain more than DNA-only models.

Difficulty: moderate. Strong fit for a bioinformatics master's student who can handle omics matrices.

5. Useful organizing win: mechanism-family taxonomy instead of rare-event list

Relevant theories/laws:

theory-1842 and **theory-1856**. Theory-1842 groups ligand presentation, splice-mediated receptor turnover changes, enhancer/SV rewiring, and phosphoproteomic activation as “regulatory rewiring” rather than unrelated oddities. Peter A Jansen Theory-1856 similarly argues that oncogene-negative LUAD is not explained by one undiscovered recurrent gene, but by substitution of canonical kinase-driver logic with alternative architectures: ligand fusions, RAF/RAS-family regulatory-site mutations, RTK exon/domain skipping, enhancer/SV rewiring, and co-loss of redundant suppressors. Peter A Jansen

Why this is useful relative to current knowledge:

The utility is taxonomic. Instead of saying “we found NRG1 here, MET exon 14 there, ERBB2 enhancer rewiring somewhere else,” this makes a reusable ontology:

```
canonical hidden lesion
fusion/splice lesion
ligand-presentation lesion
structural/regulatory rewiring
suppressor-loss convergence
protein/phospho-state activation
lineage/ecosystem state
```



Student-suitable computational experiment:

Build an NLP classifier over abstracts/full texts that tags hidden-driver papers by mechanism family. Then test whether papers/cohorts that use deeper assays discover different families at different rates.

For example:

- RNA-seq-heavy studies should recover more fusions/splice events.
- WGS/SV studies should recover more structural/regulatory events.
- proteogenomic studies should recover more output-state events.
- driver-negative residual studies should increasingly invoke suppressor-loss/state explanations.

Expected result if theory is right:

The mechanism-family label should explain study-to-study differences better than individual gene names do.

Difficulty: low to moderate. This is probably the most NLP-native project in the set.

6. Riskier but potentially high-impact: lineage gates hidden-driver access

Relevant theories/laws:

theory-1881, especially the **Four-mode partition law** and **Lineage-gates-driver law**. The four-mode law divides driver-negative/driver-poor LUAD into covert canonical signaling, lineage/quiescence-plasticity, immune ecology, and metabolic-redox stress. The lineage-gates-driver law predicts that mucinous or NKX2-1-low states preferentially admit ligand-fusion or gastric-program mechanisms, stem-like/quiescent states admit low-burden driver-poor evolution, and TRU/EGFR-compatible states admit covert receptor-kinase signaling. Peter A Jansen +1

Why this is useful relative to current knowledge:

This could produce a real conceptual advance: lineage is not just a descriptive phenotype; it may constrain which hidden driver architectures are biologically accessible.

Student-suitable computational experiment:

Use public LUAD RNA-seq plus available mutation/fusion annotations:

1. Score lineage states:
 - TRU/NKX2-1-high

- mucinous/gastric/HNF4A/FOXA/AGR2/MUC5AC-like
 - KAC/alveolar-intermediate/stem-like
 - immune/metabolic states
2. Compare hidden-driver family enrichment across these lineage states.
 3. Use literature extraction to supplement rare events like NRG1 fusions, because public bulk cohorts may have too few cases.

Expected result if theory is right:

Mucinous/NKX2-1-low tumors should be enriched for ligand-fusion/ERBB-family/lineage-rewired mechanisms, while TRU-like tumors should be enriched for covert receptor-kinase signaling.

Difficulty: moderate. The main problem is low event counts. Best done as **cohort analysis + NLP meta-analysis**, not cohort analysis alone.

7. Promising but more speculative: residual state-first / metabolic-ecosystem disease

Relevant theories/laws:

theory-1883 and **theory-1874**. Theory-1883 proposes that after hidden canonical and suppressor-convergent cases are removed, a residual class is driven more by cell-state dysregulation than a single dominant genomic driver, including quiescent/stem-like, KAC-like, NKX2-1-loss mucinous, MIF-HIF-1α glycolytic, and estrogen/SRC/GPER1-like states. Peter A Jansen Theory-1874's **Metabolic persistence law** says that when canonical oncogene addiction is weak or unresolved, LUAD persistence increasingly depends on stress-adaptive metabolic programs such as glycolysis, cholesterol handling, OXPHOS, ferroptosis resistance, and ER-stress buffering. Peter A Jansen

Why this is useful relative to current knowledge:

This is conceptually interesting, but harder to prove computationally because many ecosystem/metabolic signatures are correlative. The information gain would be in showing that the deepest residual negatives are **not random leftovers**, but enriched for recurring state programs.

Student-suitable computational experiment:

After removing all canonical/hidden/suppressor-convergent cases you can identify, test whether remaining tumors are enriched for:

- glycolysis / MIF-HIF1A programs
- ferroptosis-resistance signatures
- OXPHOS/lipid metabolism
- ER-stress/UPR
- fibroblast/CAF/endothelial/angiogenesis scores
- immune exclusion / immune-hot states

Compare these against matched canonical-driver-positive LUADs.

Expected result if theory is right:

The residual class should show stronger state/ecosystem structure than expected by chance, even after known driver and suppressor-loss explanations are removed.

Difficulty: moderate; interpretation is the hard part.

My recommended master's-project ranking

Rank	Project	Main theory IDs	Why I'd prioritize it
1	Assay-depth rescue meta-analysis	1875	Cleanest NLP+bioinformatics fit; clarifies a major confounder; low cost; likely publishable as a methods/resource paper.
2	State-first classifier for driver-negative LUAD	1841, 1881	Highest information gain if it works; directly tests whether "driver-negative" should be replaced by state labels.
3	Suppressor-loss / broken-brakes subtype test	1876	Most mechanistically crisp; easy enrichment tests; clear biological story.
4	DNA-negative but phospho-output-positive tumors	1846	Strong multiomics angle; makes CPTAC useful; tests a concrete claim.
5	Mechanism-family ontology via NLP	1842, 1856, 1875	Best NLP-native contribution; can produce a reusable taxonomy and annotated corpus.
6	Lineage-gates-driver access	1881, 1883	Conceptually strong but more limited by rare event counts.
7	Metabolic/ecosystem residual state analysis	1874, 1883	Interesting, but most vulnerable to "correlation not causation."

The one I'd do first

I'd start with a **two-part project**:

Part A: Literature/NLP assay-depth rescue map

Test **theory-1875**: how much of “driver-negative” disappears as assays deepen?

Part B: Public-data state classifier

Test **theory-1841 / 1881**: after rescue/strict filtering, do residual tumors organize into reproducible state classes?

Together, that gives a strong story:

“Driver-negative LUAD is not one biological entity. First, many cases are assay-negative but biology-positive. Second, the strict residual class is better described by recurrent pathway/state modes than by a residual negative label.”

That is probably the largest practical information gain in the file.

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It seems that "driver negative" cancer is likely not a single phenomena, but a cluster of different phenomena that together work to explain these cases. Is there a theory or theories that try to compose an overarching "theory of driver negative cancer" that explains a large part of the variance? If not, are there likely useful components here? If one thinks of driver negative cancer as a bag of M&Ms, where each color represents a different type that we don't know about, are there clusters of these theories that together are likely to explain as many "colors of M&Ms" as possible for this question?

Thought for 25s >

Yes — there **are** a few theories in the file that are trying to be overarching “theories of driver-negative cancer,” but none of them alone is quite sufficient. The best answer is that the file contains **several partial umbrella theories** that can be composed into a stronger mixture model.

The strongest composite is:

Driver-negative LUAD is an assay-relative mixture category. As testing depth increases, some cases resolve into hidden canonical/near-canonical drivers; the remaining strict-negative cases split into recurrent replacement mechanisms: pathway-equivalence, regulatory-equivalence, ligand/lineage rewiring, phospho-proteomic output states, and ecosystem/metabolic/immune state attractors.

That is basically the “bag of M&Ms” theory.

The closest existing umbrella theories in the file

1. theory-1873 — Network-Replacement Theory

This is probably the cleanest overarching theory. It says that when a canonical trunk driver is absent, oncogenic control can be restored at another layer. It explicitly proposes three major replacement modules: **pathway-equivalence**, **regulatory-equivalence**, and **ecosystem-equivalence**. It also says driver-negative LUAD should be classified by the layer at which proliferative/survival control is restored. Peter A Jansen +1

This is very close to an M&M-color model:

M&M color	Replacement module	Meaning
Red	Pathway-equivalence	Same MAPK/PI3K output, different lesion architecture.
Blue	Regulatory-equivalence	Coding driver absent, but enhancer/SV/methylation/3D genome rewires expression.
Green	Ecosystem-equivalence	Fitness maintained by immune exclusion, angiogenesis, CAFs, lineage transitions, etc.

The key limitation is that **theory-1873 does not explicitly include the assay-depth / false-negative problem as the first gate**, even though that is essential.

2. theory-1875 — Hidden Lesion Hierarchy Theory

This is the best “first-stage sorting” theory. It says that many clinically driver-negative LUADs are not biologically driverless; they sit in an ordered hierarchy of hidden lesion classes: rare coding/domain variants, fusions/exon-skipping events, amplifications/SVs, noncoding enhancer/super-enhancer mechanisms, and finally a smaller residual class dominated by suppressor co-loss or pathway-state biology. Peter A Jansen

This theory is very useful because it explains why “driver-negative” depends on how hard you looked. Its **Assay-depth rescue law** says the negative pool should shrink as testing expands from hotspot

panels to DNA/RNA/structural/noncoding profiling. Peter A Jansen

In the M&M analogy, theory-1875 says:

Some “unknown-color” M&Ms are only unknown because the lighting is bad. Before inventing new colors, improve the color detector.

That makes it a necessary companion to theory-1873.

3. theory-1881 — Four-mode ecological mosaic

This one explicitly says true driver-negative LUAD is **not a single leftover category**, but a mosaic of recurring system states. It proposes four recurrent ecological modes: **covert canonical signaling**, **lineage/quiescence-plasticity**, **immune ecology**, and **metabolic-redox stress**. It also says different studies recover different fractions of these modes because negativity definitions, assay depth, and tissue context differ. Peter A Jansen

This is probably the best “M&M-color” theory for the **strict residual** group after obvious hidden drivers have been removed.

Its four colors would be:

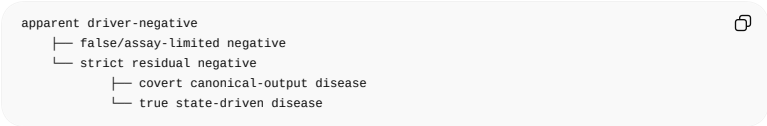
M&M color	theory-1881 mode	Meaning
Red	Covert canonical signaling	Looks driver-negative by DNA panel, but pathway output is still canonical-ish.
Orange	Lineage/quiescence-plasticity	Tumor maintained by lineage state, stemness, mucinous/NKX2-1-low/gastric-like programs.
Green	Immune ecology	Tumor behavior dominated by immune-hot/immune-cold/exclusion states.
Purple	Metabolic-redox stress	Tumor maintained by glycolysis, ferroptosis resistance, redox, OXPHOS/lipid/ER-stress programs.

The useful caveat in theory-1881 is that these are **ecological abstractions, not mutually exclusive genomic bins**. It also explicitly predicts that a deeply profiled strict-negative cohort should cluster better by these modes than by residual gene lists. Peter A Jansen

4. theory-1880 — Two-stage pathway-output framework

This one may be the most practical clinical/research framing. It says the key hidden structure is not whether a DNA driver was found, but whether the tumor has reinstated canonical oncogenic signaling output through noncanonical routes. Its novelty statement is especially useful: first correct for assay-limited false negativity, then split the strict residual class into **covert-canonical output disease** versus **true state-driven residual disease**. Peter A Jansen +1

This is the simplest overarching skeleton:



That is a good high-level model, but probably too coarse by itself.

5. theory-1874 — Convergent State Attractor Theory

This one is useful for the residual tail. It says strict driver-negative LUAD is not unified by one missing gene class, but by convergence onto stable malignant attractors: **RAS/PI3K phospho-signaling**, **lineage plasticity**, **metabolic persistence**, and **ecosystem-coupled angiogenic/immune/CAF states**. It also explicitly says it applies to the residual population after broad genomic testing and acknowledges that many apparent negatives are later rescued by deeper sequencing. Peter A Jansen

This is close to theory-1881, but with a more “state attractor” flavor.

My suggested composite “M&M” theory

I’d compose the useful pieces like this:

A Resolution-Stratified Mosaic Theory of Driver-Negative LUAD

Step 1: Apparent-negative is not a biological class

Start with **theory-1875**. The first partition is not biology; it is **observability**.

Color	Class	Main theories	What it means
Gray	Assay-limited false negative	1875, 1880	The tumor has a conventional or near-conventional driver, but the assay missed it because of panel scope, low purity, RNA/fusion detection failure, structural complexity, etc.
Red	Rare/near-canonical hidden driver	1875, 1856, 1882	Not a common EGFR/KRAS/ALK-like call, but still a rare coding/fusion/splice/amplification driver restoring RTK/RAS/RAF output.
Blue	Structural/regulatory hidden driver	1873, 1875, 1856	Enhancer hijacking, super-enhancer formation, methylation, 3D-genome/SV rewiring, outlier expression.

This layer probably explains a large fraction of routine “driver-negative” cases, especially if the original definition was narrow.

Step 2: Strict residual negatives are replacement-layer diseases

Then use **theory-1873** and **theory-1881**.

Color	Class	Main theories/laws	What it means
Yellow	Pathway-equivalence / suppressor-loss substitute driver	1873 Pathway-equivalence law; 1876 Suppressor-convergence substitute-driver law; 1878 Convergent-pathway substitution law	The accelerator is not obviously stuck down; instead the brakes are broken. NF1/RASA1/PTEN-like loss or rare pathway-member changes recreate MAPK/PI3K output.
Orange	Proteomic/phospho-output disease	1874 Convergent phospho-state law; 1880 pathway-output framing	DNA looks quiet, but phosphoproteomics/RPPA/protein state shows MAPK, PI3K/mTOR, CDK, ERBB, SRC, or other active output.
Green	Lineage-gated hidden-driver disease	1881 Lineage-gates-driver law; 1856 Driver-substitution theory	Mucinous/NKX2-1-low/gastric/TRU/stem-like context changes which hidden-driver mechanisms are accessible or selected.
Purple	Ecosystem-equivalence disease	1873 Ecosystem-equivalence module; 1874 ecosystem-coupled attractor; 1881 immune ecology mode	Tumor fitness is maintained by immune exclusion, angiogenesis, CAFs, stromal niches, or tissue-state architecture.
Brown	Metabolic-redox residual state	1874 metabolic persistence; 1881 metabolic-redox stress mode	Driver-negative cases maintained by glycolysis, OXPHOS/lipid state, ferroptosis resistance, redox/ER-stress buffering, MIF/HIF-like programs.
Pink	Lineage/quiescence/stemness residual state	1881 lineage/quiescence-plasticity; 1874 lineage-plasticity attractor	The tumor may be less “one lesion → one pathway” and more a stable epithelial state, quiescent/stem-like program, or precursor-like attractor.

That composite likely covers the most “colors of M&Ms.”

Which theories are most useful as components?

If I had to pick the **minimum set** that explains the most variance, I would use:

- theory-1875** — because it explains the assay-depth/observability problem.
- theory-1873** — because it gives the cleanest replacement-layer ontology: pathway, regulatory, ecosystem.
- theory-1876** — because it gives the strongest concrete “broken brakes” mechanism.
- theory-1874** — because it handles phospho-state and attractor-state residual disease.
- theory-1881** — because it gives the clearest M&M-color taxonomy for true residual negatives: covert canonical, lineage/quiescence, immune ecology, metabolic-redox.

That set gives you a strong overarching framework without needing all 40-ish theories.

What I would not do

I would not try to make **one single mechanistic theory** of driver-negative cancer. That would probably be wrong. The evidence in the file points toward a **mixture model**, not a unifying molecular mechanism.

A good paper-level claim would be:

“Driver-negative LUAD is not a molecular subtype. It is an assay-relative mixture category whose members become interpretable when stratified by observability depth and by the biological layer that restores malignant fitness.”

That sentence is basically the synthesis of **1875 + 1873 + 1881/1874/1880**.

Most useful “M&M color” clustering

If your goal is to explain as many unknown colors as possible, I'd use this practical seven-color model:

Color	Proposed class	Best theory IDs	Why it matters
Gray	Technical/assay-limited false negative	1875, 1880	Prevents overinterpreting weakly tested negatives.

Color	Proposed class	Best theory IDs	Why it matters
Red	Hidden canonical/near-canonical driver	1875, 1856, 1882	Captures rare coding, fusion, exon-skipping, amplification, RTK/RAS/RAF-axis events.
Yellow	Suppressor-loss / broken-brakes substitute driver	1876, 1873, 1878	Captures NF1/RASA1/PTEN-like pathway reinstatement.
Blue	Regulatory/SV/enhancer hidden driver	1873, 1875, 1856	Captures noncoding and 3D-genome mechanisms invisible to hotspot testing.
Orange	Protein/phospho-output positive	1874, 1880, 1846	Captures cases where pathway state outruns DNA annotation.
Green	Lineage-gated / mucinous / plasticity-driven	1881, 1874, 1856	Captures histology and lineage contexts that bias hidden mechanism access.
Purple	Ecosystem/metabolic residual state	1881, 1874, 1873	Captures immune, stromal, angiogenic, metabolic, ferroptosis/redox states.

The most important point is that these colors are **not necessarily mutually exclusive**. A mucinous/NKX2-1-low tumor might also have an NRG1 fusion; a suppressor-loss tumor might also have a metabolic-redox state; a regulatory/SV tumor might present mainly as a phosphoproteomic-output tumor. So the right model is probably **multi-label**, not single-label.

Best testable formulation

A compact falsifiable version would be:

In deeply profiled apparent driver-negative LUAD, most cases can be assigned to one or more of seven classes: assay-limited false negative, hidden canonical/near-canonical driver, suppressor-loss pathway-equivalence, regulatory/SV-equivalence, phospho-output-positive, lineage-gated plasticity, and ecosystem/metabolic residual state. These classes should explain more variance in pathway activity, prognosis, histology, and therapeutic vulnerability than the binary label “driver-negative.”

That is the version I would actually operationalize computationally.